

8. Practice Session

Saturday, 21 September 2024

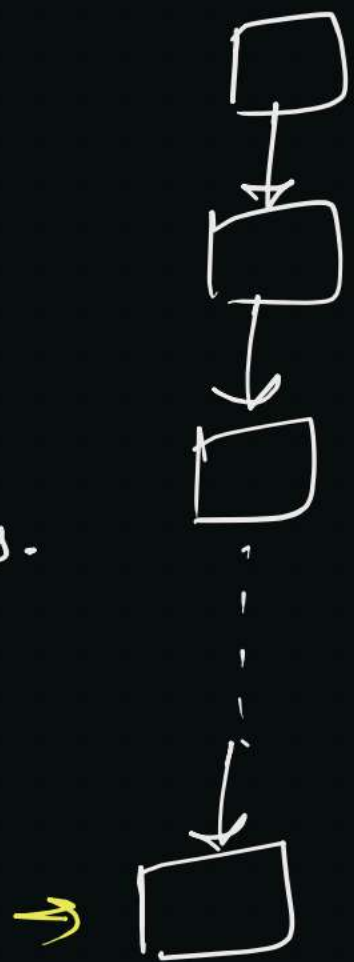
2:11 PM

① Describe a state space in which iterative deepening search performs much worse than DFS.

DFS $\rightarrow$ n searches.
 $O(m)$ searches.

Iterative deepening. $\rightarrow$ $O(n^2)$ searches.

$\frac{n(n+1)}{2}$ searches.



$b=1$

depth = $n-1$
 $n = \text{no. of nodes.}$

(2) A 3-foot tall monkey is in a room where some bananas are suspended from the 8-foot ceiling. He would like to get these bananas. The room also contains 2 stackable, movable & climbable 3-foot high crates.

↪ upstacking / downstacking.

Actions → ?

goal state → ?

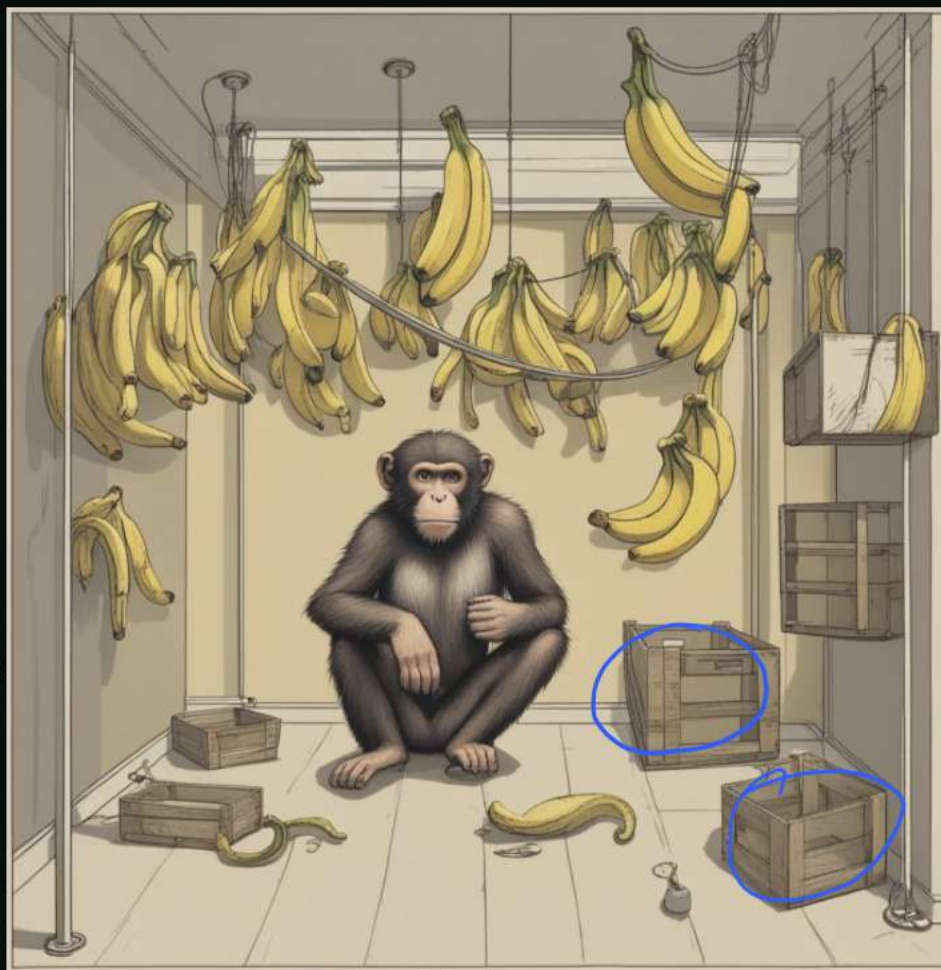
Sol.

Initial State →

Actions → { walking, shifting crates, stacking the crates, unstacking the crates, climbing up on the crates, climbing down the crates, grabbing the bananas, eating them. }

↪ final actions

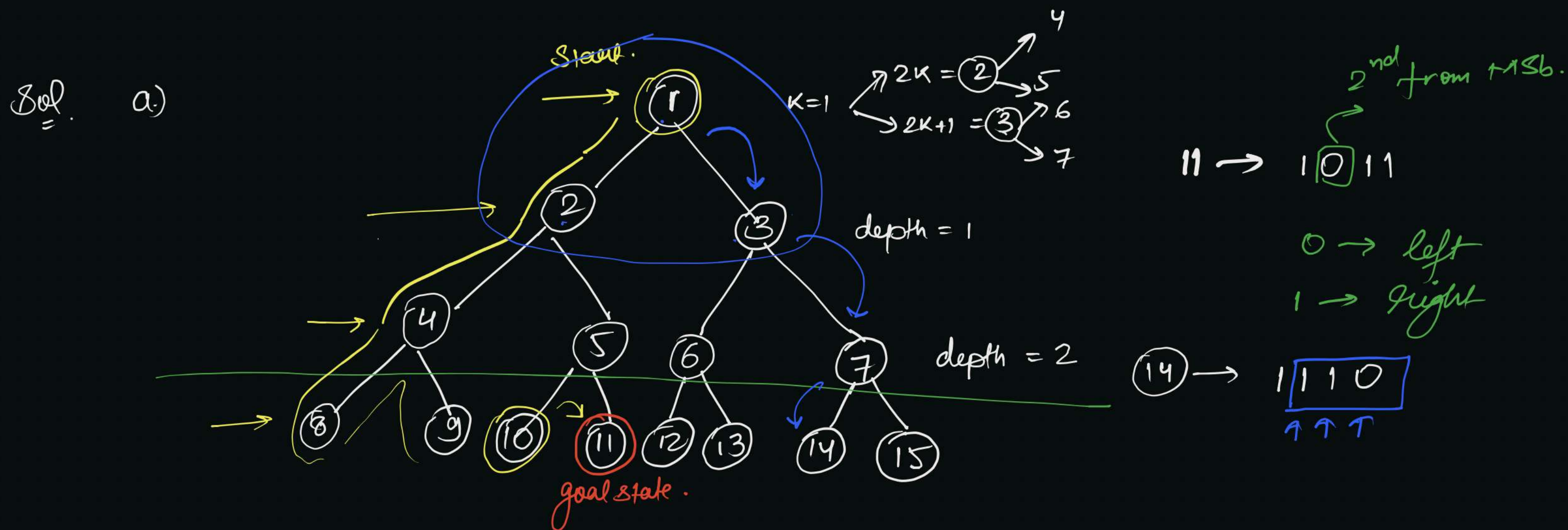
goal state → { ^{grabbing the} eating the bananas. } to reach the goal state.



8 foot.

③ Consider a state space where the start state is numbered ① & each state k has 2 successors : $2k$ & $2k+1$.

- Draw the portion from states 1-15.
- Suppose the goal state is 11. List out the ordering of nodes using:-
 - BFS
 - Depth limited ($l=2, l=3$) nodes
 - Iterative deepening. $\rightarrow$ (# searched)
- Call the action of going from state $k \rightarrow 2k$ as left & $k \rightarrow 2k+1$ as right. Can you find an algorithm that outputs the sol/path to this problem without any search?

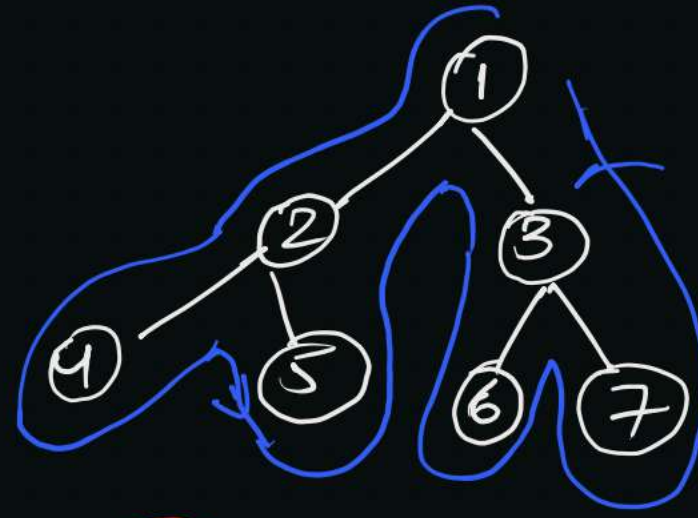


(i) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 10 \rightarrow \textcircled{11} \rightarrow \text{goal state.}$

(ii) $\text{limit} = 2$

$1 \rightarrow 2 \rightarrow 4 \rightarrow 5 \rightarrow 3 \rightarrow 6 \rightarrow 7.$

$\text{limit} = 3$



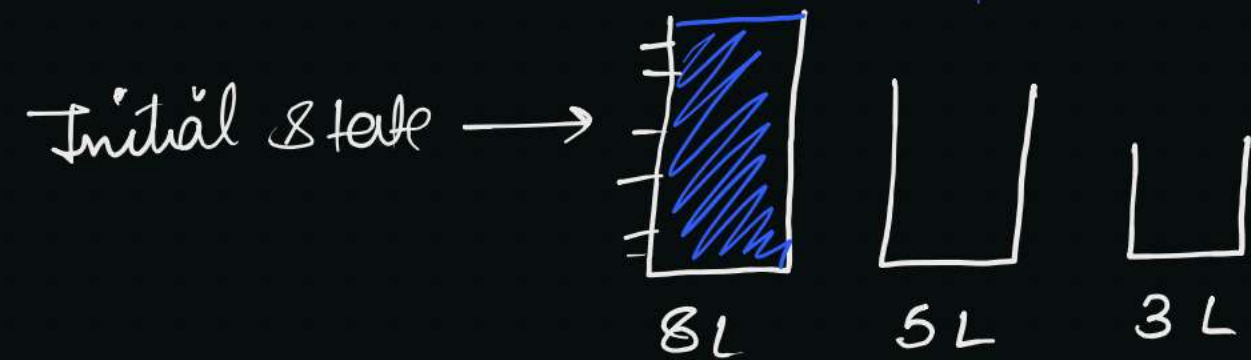
$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 9 \rightarrow 5 \rightarrow 10 \rightarrow \textcircled{11}.$ $\rightarrow \text{goal state.}$

(iii) Iterative deepening :

1, 1-2-3, 1-2-4-5-3-6-7, 1-2-4-8-9-5-10-11
 $\textcircled{1}$ $\textcircled{3}$ $\textcircled{7}$ $\textcircled{8}$

nodes searched = $1 + 3 + 7 + 8$
= $\textcircled{19}$

- (14.) You have 3 jugs with capacity 8, 5 & 3 L. (Water Jug problem)
 8L jug $\rightarrow$ filled with water & 5 & 3 L are empty.
 Need to measure 4L of water.



(8,0,0)

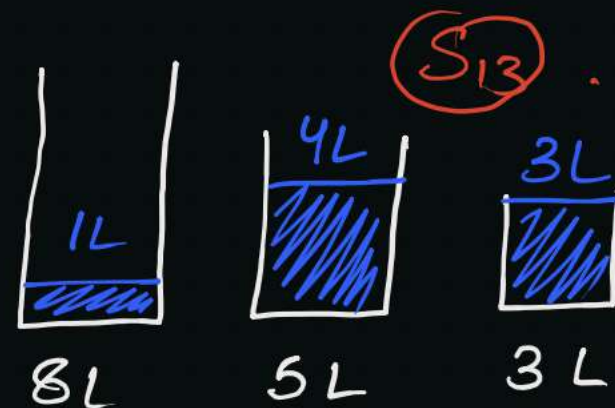
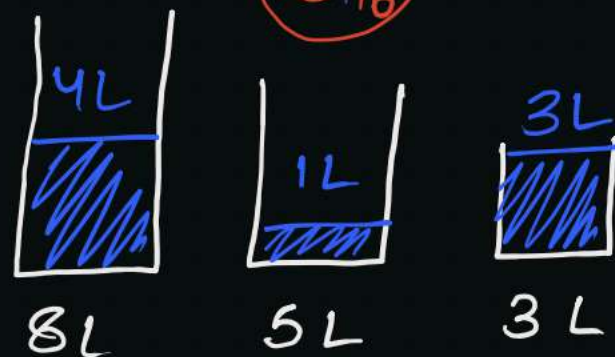
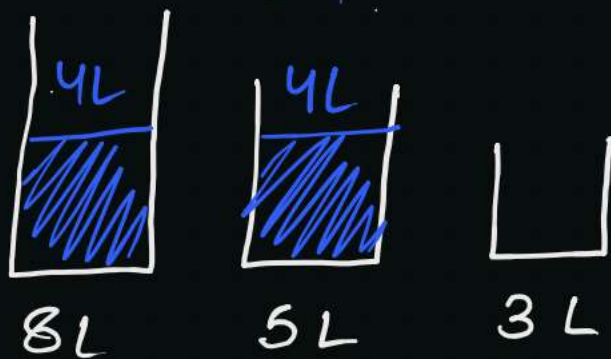
Give T/F to these statements.

- (i) the state space graph has 10 states.
- (ii) 7L measurement is not possible.
- (iii) there can be 2 different ways in which 6L of water can be measured.

(iv) 5 ways to measure 4L of water.

(F) (v) the state space graph will be having only reversible edges. (bidirectional)

Goal States $\rightarrow$



⑤ Select the best search strategy

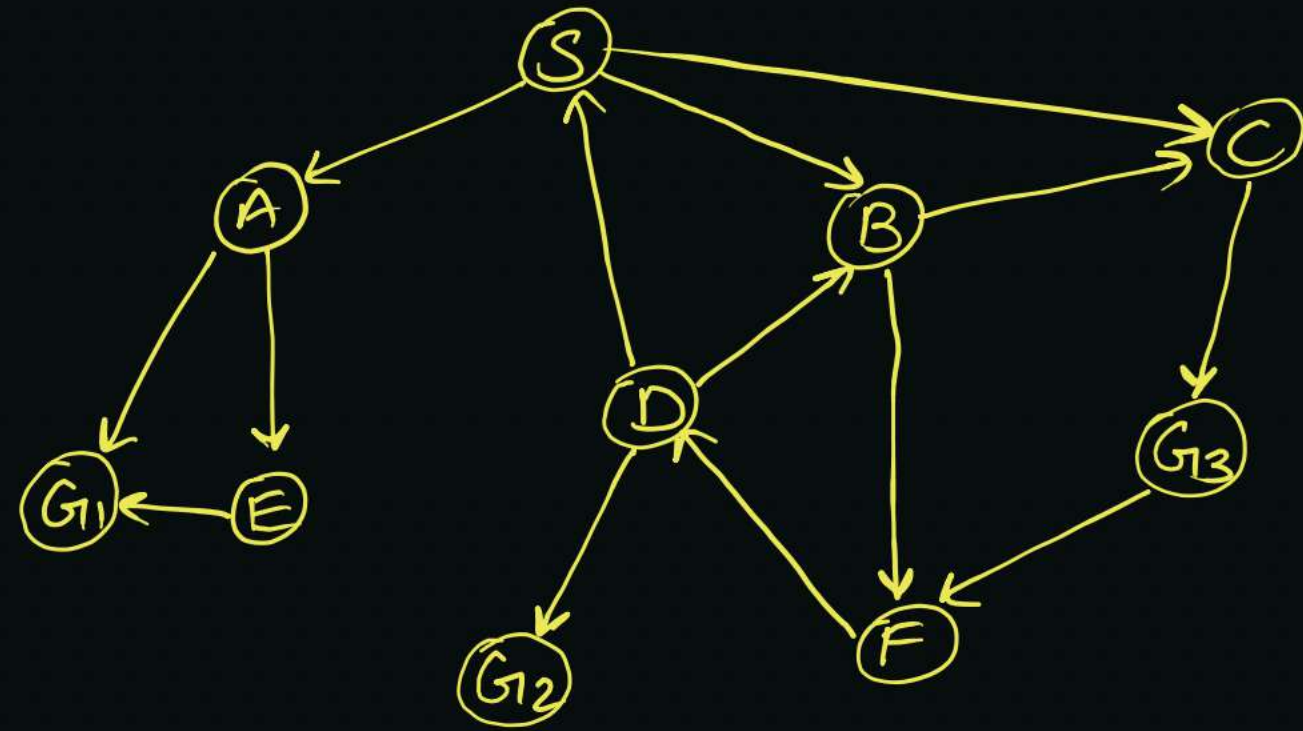
a.) All the actions are equal cost. Time is not a problem & have plenty of space. But you must find the optimal solution.

b.) The search space is large & depth of the solution is not known. Cost information is not given. You need to find the shallowest solution (optimal) || but don't have enough memory.
[only due to this]
Constraint

→ Iterative deepening

≈ bidirectional bfs →

6.



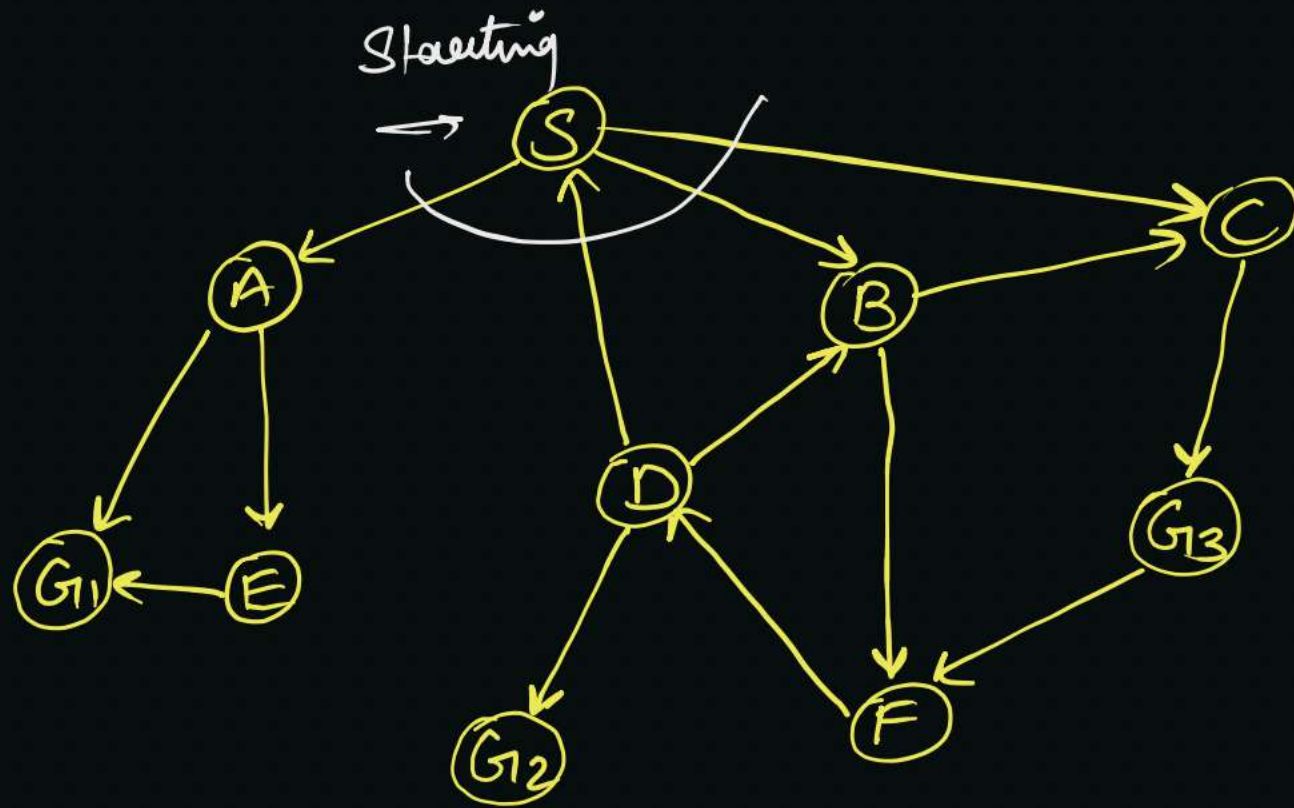
Consider this state space graph, with S as starting node & G_1, G_2 & G_3 are the goal states.

- What is the order in which the nodes are removed from the fringe. $\rightarrow$ (Iterative - deepening)
- Which goal state is reached first? (G_1) . ✓

Sol.

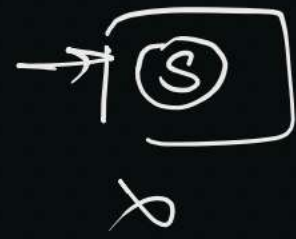
S, S-A-B-C,
S-A-E-G₁

node
searches = 9



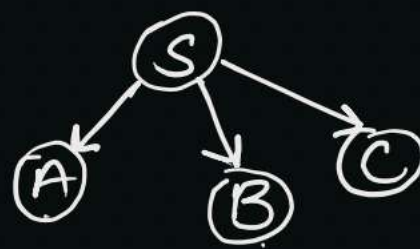
goals = {G₁, G₂, G₃}

I - Iteration

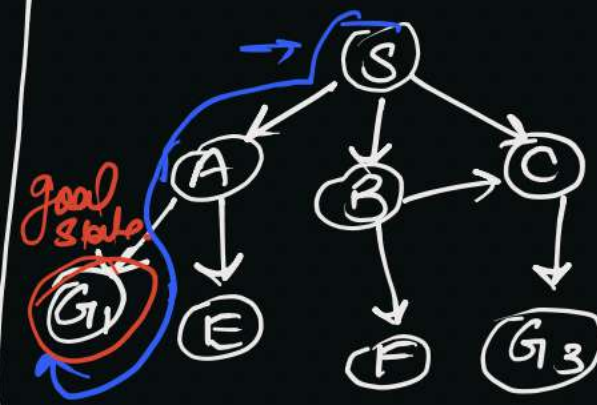


we are not
expanding S

II - Iteration



III - Iteration



S → (d+1) searches,
S → d → expanded.

Q7. Let b be the branching factor of the search tree. If the optimal goal is reached after d actions from the initial state, in worst case, how many times will the initial state be expanded for IDDFS?

GATE, DA
Sample paper
2024

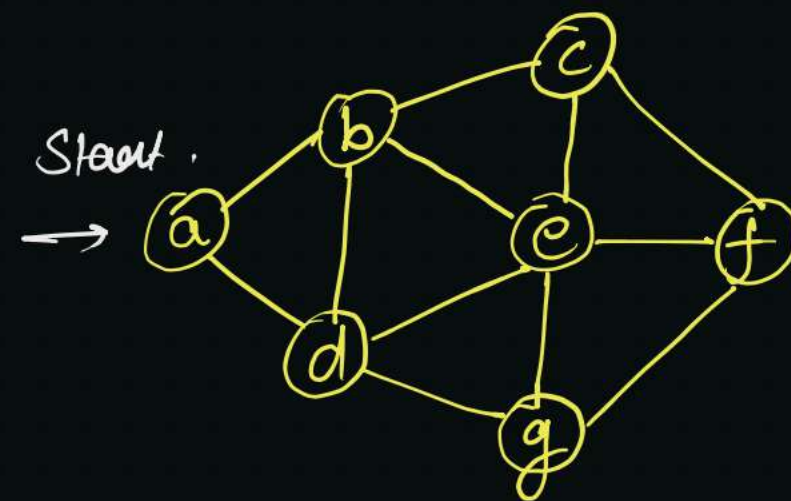
a) $d-1$ c) $d+1$
~~b) d~~ d) b^d

$(d+1) \times$

Q8. Consider the following sequence of nodes for the undirected graph given below.

GATE
IT, 08

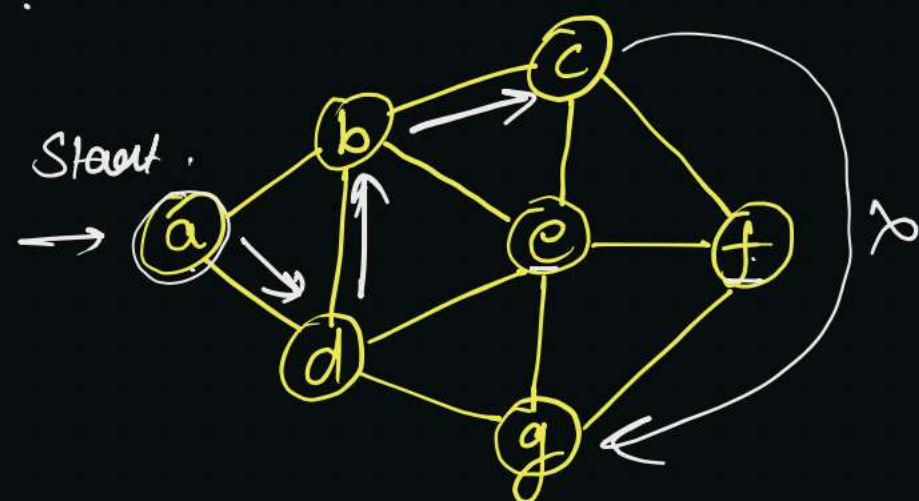
1. a, b, e, f, d, g, c ✗
- ~~2. a, b, e, f, c, g, d~~
- ~~3. a, d, g, e, b, c, f~~
4. a, d, b, c, g, e, f ✗



Performing DFS from node a , which sequence(s) are possible?

Sol.

1. $a, b, e, \boxed{f, d}, g, c$ ✗
- ~~2. a, b, e, f, c, g, d~~
- ~~3. a, d, g, e, b, c, f~~
4. $a, d, b, \boxed{c, g}, e, f$ ✗



(H.W.) The MGLC problem.

Q. A man needs to transport a lion, a goat & a cabbage across a river. He has a boat in which he can take one of them at a time. It's only his presence that prevent the lion eating the goat, & the goat from eating the cabbage. How can he take them all to cross the river?